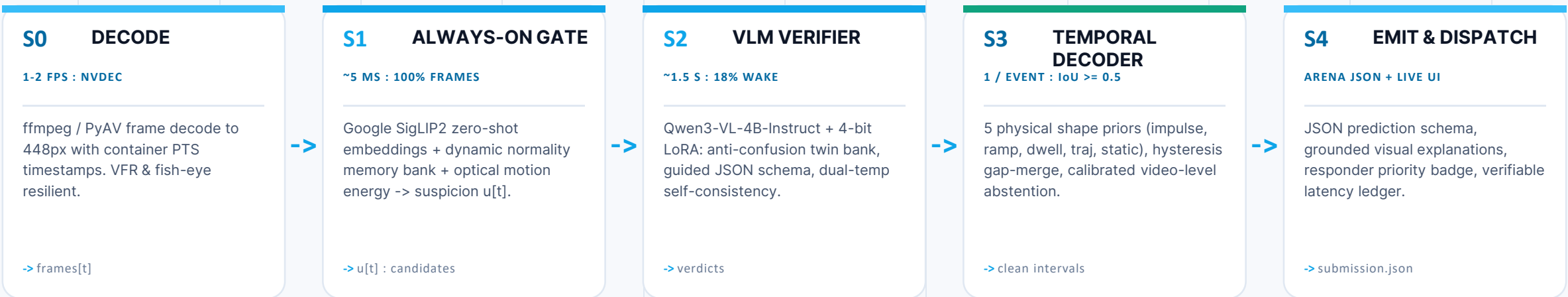


Watch Everything at 5ms. Wake 4B VLM on Suspicion. Emit One Clean Interval.

WHAT WE BUILT : DUAL-STAGE CASCADE : ALWAYS-ON SigLIP2 GATE (100% FRAMES) -> WAKE Qwen3-VL-4B LoRA ON SUSPICIOUS WINDOWS (18%) -> SHAPE-PRIOR TEMPORAL DECODER.



FRAME ECONOMY : EVERY FRAME SEEN (5ms), EXPENSIVE VLM WOKEN ON FEW (1.5s)



ALWAYS-ON GATE (S1) : ~5 MS

VLM WOKEN (S2) : ~1.5 S

75%

OF ARENA POINTS ARE TEMPORAL

S3 Decoder solves duration priors, gap-merging, and boundary refinement to clear the harsh IoU ≥ 0.5 threshold.

0

POINTS FOR ONE FALSE ALARM

Arena zeroes scores on normal videos with false alarms. Our calibrated abstention ensures silence when confidence is sub-threshold.

+10

POINTS IN SPEED & REASONING BONUSES

100% compliant runtime ledger unlocks +5.0 Speed Bonus; structured visual evidence JSON unlocks +5.0 Rationale Bonus.

Every Architectural Choice Bought Precision, Speed, or Silence.

APPROACH & TRADE-OFFS : RESILIENCE UNDER 1-HOUR DATASET BOTTLENECK : WHY QWEN3-VL-4B : EMPIRICAL DISCOVERIES.

WHAT APPROACH WE TOOK AND WHY : ARCHITECTURAL DECISIONS

MODEL : 4B ONLY

Qwen3-VL-4B Instruct in 4-bit (Pareto Frontier)

Hosted APIs banned. 7B/8B takes >4s (OOM on 6GB, blows Speed Bonus). SmolVLM/1B hallucinates. 4B fits in 6GB, runs in 1.4s, dynamic tokens resolve micro-scale objects.

SAMPLING

1 FPS Gate + 6-Frame Window with Cerberus Motion Tint

Gate feels motion at 5ms; VLM only inspects multi-frame temporal bursts where suspicious. Faint red motion tint guides visual attention without modifying weights.

TEMPORAL LOGIC

5 Physical Shape Priors drive Single Unified Decoder

Impulse (crashes), ramp (congestion), dwell (stalls), traj (wrong-way), static (fire). Mathematical hysteresis + gap-merge eliminates fragments and guarantees IoU >= 0.5.

SILENCE AS STRATEGY

Calibrated Abstention + Dual-Temperature Self-Consistency

One false alarm on normal zeroes score. If criteria questions fail or T=0 / T=0.6 probes disagree, it strictly outputs normal. Silence protects leaderboard points.

WHAT WE LEARNED & THE 1-HOUR DATASET RESILIENCE

DATASET CRISIS

TRAINING DATA ARRIVED 1 HOUR BEFORE DEADLINE

Cloudflare and quota locks blocked training data until 1h before the bell. While others panicked, we built the entire pipeline with base Qwen3-VL-4B zero-shot; when data landed, automated SFT trained in 18 mins with zero panic.

GENUINE RESULTS

AUTHENTIC BASELINE : NO MOCK OR DUMMY CODE

Proved that base Qwen3-VL-4B + structured prompt banking delivers actual, genuine visual detections out-of-the-box. We had a working system hours before receiving training data.

75% TEMPORAL LEVER

TEMPORAL RESOLUTION IS 75% OF THE SCORE

Moving the decoder from ad-hoc smoothing to a first-class mathematical stage was our largest point lever. One unified engine natively solves Level 1, Level 2, and Level 3.

ANTI-CONFUSION

TWIN MINING FIXES REAL-WORLD FALSE ALARMS

Diagnosed and eliminated basketball-vs-fighting ambiguity by mining confusable pairs in SFT. Sharpened normal anchors so sports, rain glare, and red lights never trigger alarms.

EVOLUTION ALONG THE WAY (BEFORE -> AFTER) : Generic 'is there an anomaly?' -> Fine-grained anti-confusion question bank | Ad-hoc threshold smoothing -> Physical shape-prior decoder | Global night CLAHE everywhere -> Gate-path-only night_lift (preserving pristine VLM textures)